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"E.C.H.O Efficient Contextualized History Orchestrator"

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CERTIFICATE

This is to certify that the dissertation entitled

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DECLARATION

We hereby declare that the Capstone Project Phase - 1 entitled **"E.C Efficient Contextualized History Orchestrator"** has been carried out by us under the guidance of Dr.M.Shalima Sulthana**, Associate Professor** and submitted in partial fulfillment of the course requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester Aug. – Dec 2025. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ABSTRACT

In day to day digital work, most people move between several applications such as Chrome, Gmail and YouTube. We often view something important while browsing or while watching a video, but after a few days, it becomes difficult to remember where exactly we saw it. Usually, instead of searching back through mails or history tabs, we simply start the search again. Because of this common behaviour, we thought of building a system that can remember these digital traces in a meaningful manner and help the user retrieve them when needed.

Echo is designed to act as a simple digital memory rather than just a history collector. Instead of only saving the URL or the mail subject, it looks at what the user was probably trying to do at that time and relates the activities. For example, if someone searched online, later watched a tutorial, and then received an email about the same topic, Echo tries to link all three together.

The system is mainly developed to run inside the Chrome browser through an extension and can optionally use cloud support if required. Storage preference is left to the user, allowing them to keep their data local by default. Basic analytics and simple visuals are included so that users can observe how their time is being spent online.

As digital content keeps increasing each year, recalling previous interactions has become a challenge. A tool like Echo can reduce this effort by helping users refer back to what they engaged with earlier. In future, the system can be enhanced to support more sources and provide deeper insights based on user behaviour.

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CHAPTER 1
INTRODUCTION

1.1 Project Overview

We generate vast amounts of data which we often struggle to maintain. It's a lot of information to remember our past interactions and understand how we have spent our time online. As a result, personal knowledge becomes fragmented and difficult to search.

With rapidly increasing digital communication and information consumption, users need to frequently experience cognitive overload. Important e-mails are lost, browsing sessions are lost, video content is difficult to recall again and again. Traditional search doesn't offer engines that combine multiple users, multiple apps together to retrieve data. With Echo, we use large language models for making more intelligent organization of personal data through which data can be retrieved better.

Our project ECHO proposes a unified personal assistant based on AI to retrieve information in Gmail, Youtube and Chrome history. This combines data aggregation with Retrieval Augmented Generation (RAG). ECHO allows users to view their own digital footprint in natural language and receive personalized answers.

1.2 Context and Motivation

With rapidly increasing digital communication and information consumption, users need to frequently experience cognitive overload. Important e-mails are lost, browsing sessions are lost, video content is difficult to recall again and again. Traditional search doesn't offer engines that combine multiple users, multiple apps together to retrieve data. With Echo, we use large language models for making more intelligent organization of personal data

1.3 Project Scope

.The scope of this project is vast. It covers:

- Integration of Gmail API, YouTube API, and Chrome history API together
- Local processing of data from the initial phases of inquiry to ensure the privacy is there
- Building vector-based search memory through embedding
- A Chrome extension that is a proper chat interface for the users
- Features like summarization, keyword search, time spent, and digital well-being

1.4 Project Objectives

ECHO's main objectives include:

Developing a system capable of retrieving, organizing, and summarizing data of users

Implementing a RAC pipeline that allows querying of personal data in natural, understandable language

Designing an interface for a Chrome extension allowing seamless interaction with the system

Promoting digital welbeing through Information and time spent in watching videos and browsing the internet.

Improving personalization by analyzing the behavior of the user and adapting to their responses accordingly.

1.5 Problem FrameworkReports organized is:

Users currently have interaction with various online services, generating their own isolated data sets. Emails, videos, and web history exist in separate columns, requiring users to manually switch between those platforms to locate information. This leads to a lot of time waste, difficulty in finding the data, and difficulty in recalling what was done before. Echo is solving this problem by acting as a centralized memory layer capable of intelligently searching and summarizing the data. 1.5 Problem Framework

1.6 Report Structure

Chapter One: follows the presentation and introduction of scope of the project

Chapter Two: defines the problem in detail and describes the limitations of other systems

Chapter Three: is the literature survey which searches the literature on rack pipeline personalization and digital well-being

Chapter Four: are the results of Phase I and what we have learned from the Phase I project

Chapter Five: is the implementation of Phase II and what we are going to do in the future

CHAPTER 2

PROBLEM DEFINITION

2.1 Problem Statement

Users are generating large amounts of personal data in Gmail, YouTube and Chrome in their browsing histories. Information is scattered amongst various platforms making it difficult to retrieve, search and analyze what they are doing. Tools that are already existing perform only on the only on specific APIs like either YouTube, Gmail or Chrome not all of them together and they also include screenshots and video recording not using API ingestions

Unified system is needed to organize this support natural language and provide summaries and other useful information.

2.2 Project Need

This project is useful because:

Personal data is very fragmented across various platforms and not available in a single location.

Tools that are already existing require exact keywords, which a user may not remember.

A unified tool capable of simultaneously searching Gmail, YouTube, and Chrome doesn't exist.

Users are facing various challenges in digital overload and need smart summaries to save their time.

AI-powered personal assistants are increasing for productivity and for well-being. and the RAG pipelines.

2.3 Limitations of Existing Systems

Existing tools are not fully solving the problem, but there are very significant limitations in integrating API personalization and retrieving informationSummary of the gaps:

Product / Tool	What It Does	Limitations
Google Gemini / Bard	Can summarize Gmail when granted permission; can answer AI queries	Cannot access Chrome history; cannot combine Gmail + YouTube + browsing; no unified memory layer; no persistent personal database; no personalized analytics.
Microsoft Recall	Screenshots of PC activity for personal memory	Works only on Copilot+ hardware; privacy concerns; no Gmail API or YouTube API; only captures screen, not structured semantic data.
Rewind AI	Local on-device memory (records screen/audio)	Doesn't truly integrate Gmail or YouTube via API; relies on screen capture; not cross-platform; heavy resource usage; no Chrome-extension chat interface.

Product / Tool	What It Does	Limitations
Mem AI	Personal AI note-taking and knowledge base	Requires manual input; no automatic Gmail/YouTube/History ingestion; not optimized for personal digital footprint analysis.
Comet / Atlas	Provide integrations with Gmail, Calendar, YouTube, and some browsing metadata	Integrations are not deeply semantic: they sync raw data but do not support RAG-based natural-language querying across combined sources. No Chrome-level browsing history ingestion. No unified cross-source analytics.
Google Takeout	Allows exporting Gmail/YouTube/Drive data manually	Not automated; not real-time; cannot answer natural-language questions; no analytics or personalization.

Table 2.1: Comparison of Existing Systems

Summary of Gaps: Existing systems fail because none of them offer:

Existing systems are failing because they don't offer integration between sources such as Gmail, Chrome, and YouTube personal searches.

Semantic-based chat interface in Chrome extension, intelligent analytics like insights on time viewing, email summaries, digital well-being, etc.

Option that prioritizes privacy on a low cost.

2.4 Description of the Proposed Solution (High Level)

Echo proposes an integrated AI system designed to unify personal data from mail, YouTube, and Chrome browsing history in a single intelligent platform. This system enables users to explore the digital footprint using natural language. The solution is built through modular components that ensure scalability, intelligence, and privacy. The high-level solution is shown below.

2.4.1 Multi-Source Data Acquisition Layer

Echo gathers real-time and historical data from multiple ecosystems using authorized APIs and browser-level mechanisms.

- Gmail API extracts email threads metadata and message content for semantic retrieval.

- Chrome History API captures browsing events, visited URLs, page titles, queries, dwell time to model user behavior.

- YouTube API retrieves watch history metadata of videos and user interaction signals.

2.4.2 Data Preprocessing and Normalization Pipeline

Because each data source is heterogeneous, Echo employs dedicated preprocessing workflows which include:

1. Text cleaning and tokenization
2. Removing noise, HTML, and unnecessary formatting
3. Metadata normalization (aligning time stamps, sanitizing URLs, and mapping conversation threads)
4. Standardizing identifiers
5. Semantic standardization (converting mail video titles and web page content into a uniform representation)

.

2.4.3 Generation of semantic embeddings

After finishing preprocessing, Echo converts clean data into high-dimensional vector embeddings using transformer-based models. This enables semantic search across mails/videos/browsing history, alignment and comparison of information from unrelated platforms, clustering and pattern discovery. These embeddings are stored in high-performance vector database

2.4.4 Local or Cloud-Based Vector Database

Echo is using Vector database (such as FAISS, Chroma, and Milvus) to store embeddings. The database supports fast KNN similarity search, stores long-term memory of Gmail, YouTube, and Chrome, enables cross-platform retrieval within seconds, and preserves privacy when deployed locally. The Vector database is the core memory structure for Echo

2.4.5 Retrieval Augmented Generation (RAG) Pipeline

The Rack Engine processes queries and produces personal responses in three steps:

1. Query understanding: interprets what the user's intent is
2. Document retrieval: users' vector similarity to fetch relevant data from mails, history, and YouTube
3. LLM reasoning: a local cloud-based LLM synthesizes retrieved data into coherent, factual answers.

This ensures accuracy, grounding, and contextual relevance

2.4.6 AI-Driven Summarization and Insight Generation

Beyond Rhetorical Echo generates meaningful insights such as:

1. Summaries of unread or important mails
2. YouTube Watch Time Analysis and Digital Wellbeing
3. Trending Browsing Themes
4. Content Recommendations which are related to previous ones
5. Daily or Weekly Digital Behavior Summary

These insights make Echo proactive and insight driven

2.4.7 Chrome Extension Interface for Natural Language Interaction

Users can interact with Echo through a lightweight Chrome extension that provides:-

1. A proper chat interface
2. Smart notifications and personalized suggestions
3. Speedy and efficient browsing history access
4. Retrieval of data without switching apps
5. Smooth and interactive communication

2.4.8 Privacy, Security, and Local Infrastructure

Privacy and protection is Echo's Number one priority

1. Personal data remains fully on device
2. Vector storage facilitates long term memory
3. Open source LLMs are used so no cloud models accessing data
4. Minimum and fully disclosed permissions

2.4.9 Scalability and Modularity

The system is modular, meaning:

1. Adding new data sources is fairly simple
2. Replacing LLMs or vector engines accordingly
3. Each component is scaled independently
4. Future proof and adaptable with various needs with flexible UI expansion

2.5 Echo's Unique Selling Points (USPs):

2.5.1 Enhanced Multi-Source Retrieval (RAG)

Gmail,Youtube and Chrome histories are in one single vector memory which enables queries like

“Show me what videos i watched on the day i received this internship mail”

“Which websites and videos i saw on supervised learning”

2.5.2 Integrating Chrome history

Using Chrome API, Echo accesses browsing metadata in detail, most current competitors dont read local browsing history through API integration giving Echo an advantage in search patters and semantic searching

2.5.3 Combined Memory Queries

By unifying memory from different sources, Echo supports cross querying from the same in form of

1. Timestamps
2. Behaviour correlations
3. Semantic searches

This is higher level than just mail or video assistance

2.5.4 Real-time Digital Wellbeing Analytics

Echo generates digital wellbeing stats and insights as well, like:

- Time you're spending on YouTube
- Frequent browsing themes
- Distraction alerts
- Daily and weekly summaries
- Regret button

2.5.5 Privacy-Preserving Local Architecture

Echo supports running though open source LLMs and local vector storage ensuring no personal data is being uploaded anywhere

The costs of running it also are extremely low

Echo also gives maximum privacy control and transparency

2.5.6 Conversational Interface via Chrome Extension

Echo integrates directly with the browser as an extension-based persistent assistant. Users can access their digital memory without having to switch between tabs or use a separate application.

This highly integrated interaction model improves usability and reduces friction compared to external applications.

2.5.7 Personalized Conversational Intelligence

Echo functions not only as a data aggregator but also as a personalized semantic assistant.

It adapts its responses based on long-term behavioral patterns, offers contextual recommendations (e.g., similar videos, relevant articles), and remembers past interactions (e.g., "Last week you searched for dog bite cases; would you like a summary?"). This level of personalized and contextualized conversational intelligence is not offered by Comet or conventional AI tools.

CHAPTER 3

LITERATURE SURVEY

3.1 Academic Papers summary

1. Unified Structure & Content Search (Wang et al., CIKM)

This is about searching across different types of data by giving a method that combines search across both folders and the text within files. It is a system that can find info from different personal data types and get relevant information from these structurally different platforms.

2. Retrieval-Augmented Generation (Lewis et al., NeurIPS)

This paper is about a Retrieval-Augmented Generation (RAG) framework which is an AI method where the model first gets supporting info before replying to the query. Then it gives answers that are more accurate then without it, which is pretty useful.

3. Smart Browser History Search (Microsoft Patent)

This is for a system of smart browser history search that uses Natural Language Processing and semantic indexing so that the system can understand the meaning of the query and let users ask natural questions.

4. Human-Centred AI for Wellbeing (JMIR)

This writes of te landscape of AI-powered digital wellbeing tools by comparing existing solutions. Its concludes that AI can help users manage screen time and habits beyond simple tracking.

5. Browsing Behaviour Analysis (IJ 2019, IEEE)

This is a way for analysing user activity using Data mining on browsing logs. researchers were able

to identify and figure out patterns of focus and distraction online by using data mining.

6. PersonaRAG Contextual Agents (Li et al., arXiv)

This talks about a PersonaRAG framework where AI agents give answers according to user context that they have. the system can provide more personal recommendations that are matching to their needs by making a persona from the users data.

3.2 Analysis of the Problem Domain and Existing Solutions

1. Commercial Competitor: Rewind.ai

It uses privacy first way by recording everything a user does and storing all that on the user device, not on any external server . It is a strong and private backup of the user digital records, with some features like meeting transcription and universal search. But rewind is platform-specific and so it cant really handle cross platform all that well

2. Privacy Architecture: Local Differential Privacy (LDP)

LDP is a solid math architecture for giving privacy. It applies protection at the owner's side then sends this masked data to the processor to work on it. it gives better privacy because the server never receives the raw data from the user.

3. Technical Challenge: Data Synchronization Conflicts

There are problems in data synchronization at some fail points like data conflicts from multiple updates happening to the same data item at the same time, and schema mismatches where data structures vary between different platforms like between chrome and youtube. Because of these problems the data we get is inconsistent and terrible quality which just makes it harder to do anything with it and needs a lot more processing for it to me even somewhat usable.

3.3 Gaps Identified

- Commercial Competitor Gap: commercial tools like Rewind.ai are fairly strong but they focus on local first, desktop centric records only. They cant do multiple platform API integration and so are stuck with like 1 platform. They also focus on recall only and not using the data for analytics.
- Privacy-Analytics Gap: A gap exists in using the data from the user to make user facing dashboards that give them analytics and patterns of their online activities.
- Technical Unification Gap: joining different data sources is a big problem because of conflicts coming up from the difference in data formats used by different platforms.

Chapter 4

OVERVIEW OF DATASETS

4.1 Description

Nature: The data that echo will use is dynamic, real time, and user centric data so the data in the dataset is not static it continuously evolves as an info stream from user activity as the user does more things.

access to this data should be highly restricted and private since it often has sensitive details of the user that shouldn't be at risk of leaking. the dataset is live and constantly updating and increasing in size through user interaction and activities online. To ensure privacy and maintain compliance with data protection standards like GDPR and CCPA, all data remains local to the specific user instance.

4.2 Sources of Dataset

Data is taken in through the Chrome Extensions from three separate sources. Each gives a different dimension of the user's digital engagement:

1. **Browsing Activity:** captured using the Chrome History API and specialized scripts. This source givs a user's web interactions like the URLs they visited, page metadata of that URL, and time the user spends on them.
2. **Communication:** focuses on user communication ingested through the Gmail API. This lets the system to process email threads, subject lines, and timestamps, getting the flow of information and discussion topics.
3. **Media Consumption:** it is for media consumption captured through YouTube integration. This source checks the user's watch history, video metadata, and interactions with Shorts.

4.3 Dataset Statistics & Distribution

Volume: In terms of volume, the dataset is always increasing, so its size is never fixed but grows as the user performs more digital activities so as the user interacts with the monitored applications, dataset also becomes bigger.

Complexity: it is high-dimensional for the data. this dataset has text and structured data formats such as vector embeddings, which are needed for being able to actually use the semantic search capabilities that we are putting in, there is also detailed behavior logs used for advanced analytics that need it as well.

Scale and Purpose: The system is supposed to handle the information overload problem so this design consideration has to effectively process possibly thousands of individual interaction points per user, per week without becoming too slow to bee of any use, else it would take too long to do anything and then becomes useless for any practical applications.

CHAPTER 5

CONCLUSION OF PHASE 1

5.1 Outcomes of Phase 1

In the Capstone Phase I, we were able to lay the groundwork for Echo. The Phase I first began with finding the real world problems that the people face, not only students, not only teachers, and not only employees, but the entire world, which is wasting a lot of time in shifting tabs and endless scrolling through endless pages of browser's, YouTube's, and Gmail histories. Through numerous recent studies, we found out that about 2.5 hours daily are spent in finding the info, and 70% of what was earlier searched is forgotten within 24 hours.

To tackle this real world problem, we came up with a solution: ECHO. It not only unifies your browsing, email, and YouTube history, but also goes a step ahead and gives you better digital analytics for your better digital well being.

5.2 Feasibility Study

After finding the solution for this real world problem, we moved ahead with the feasibility study where me and my team had gone through a numerous amount of documentation of APIs, research papers, and numerous other studies to know whether the project is really feasible. After our extensive study, we learnt that there are many friendly and well documented APIs from Google, and also many student friendly plans so that we could use good databases and good LLMs. And with a timeline of 16 months, it's totally feasible.

5.3 Competitor Study

Next was our competitor study. We found out that there is no real solution that actually tackles this problem, while many just focus on a single browsing stream like either Gmail or YouTube or Browsing History, but none truly unifies them together like Echo.

While Gemini only focuses on Gmail, Microsoft Recall collects screenshots through the entire system and not only from browser, they don't actually provide the solution to these problems. While in recent months, major tech giants like OpenAI and Perplexity have launched their own browsers, but still they don't do what Echo does. They could either collect data from either browser history, Gmail, or YouTube at a time. But Echo unifies them together and gives you all at a time.

Echo also excels in better digital analytics where it allows users to identify their time not only in websites they visited but through labels such as "study", "work", "entertainment". Echo goes a step ahead while giving the users a freedom to create their own labels so that they could identify their time spent in that. Echo also gives a special novelty feature of the regret button which allows users to leave their bad browsing activity.

Thus, we can conclude that there is no real solution in today's world that solves the problem. Hence, Echo itself is a novelty product.

5.4 Literature Survey

At last, we moved to a more detailed literature survey. Through research papers from IEEE and Springer, and a few patents, we learnt the architecture and the techniques to efficiently tackle them. None provides the exact solution, but they provide us a foundation and inspiration to move forward.

5.5 Conclusion

This brought to the end of our Capstone Phase I, where we got a foundational knowledge about our product and a basic next step of architecture understanding.

Lastly, across the three reviews in the Phase I, our project idea was highly appreciated by all three panel members for our innovation to solve the real world problem. Not only that, the proof that many tech giants are working in this industry gives us pride that we were able to identify a problem that even the tech giants are solving now, and will surely give us huge experience for our future.

CHAPTER 6

System Design and Proposed Approach

When we started working on Echo, it did not come from some complex technical concept, but actually from what we noticed happening to us almost every single day. We keep shifting between different tabs, checking mails, watching videos, and then when we want to look back at something we saw earlier, we simply cannot find it. Most of the time, people end up searching from the beginning rather than going through their history. So, in this chapter, we explain how we are planning to build Echo in such a way that it behaves like a basic digital memory, rather than just another collection of history items.

The main idea behind the system design is quite straightforward. Each part of Echo should do one particular job, and still fit into the complete working structure. Even though Chrome, Gmail, and YouTube are technically separate platforms, for the user they are often part of the same activity. Echo tries to understand this user perspective and quietly joins this data in the background, while still making sure that the user fully controls what is used or stored.

6.1 Architectural Overview

The architecture is created using a modular structure. This means every module has a clear responsibility and even if we update one of them later, the remaining system continues working properly. Overall, Echo consists of three main data modules for Chrome, Gmail and YouTube, a central intelligence module for processing queries, and additional modules for analytics, privacy handling and device sync.

6.1.1 Chrome Activity Module

This module keeps track of which websites the user visited and how long the tab was active. Instead of saving only the link and time spent, Echo tries to understand what kind of session it was. For example, if someone spends a lot of time continuously on a documentation or tutorial page, it most likely indicates serious research or studying. On the other hand, switching across several sites quickly may simply show casual browsing. This kind of basic understanding helps later when summarising user activity.

6.1.2 Gmail Processing Module

Most important conversations and updates come through email threads. With the user's permission, this module analyses selected mails and finds the main points so that users don't need to go through the full conversation again. Even if the user only remembers it vaguely, like "the mail about the project submission", Echo should still be able to retrieve it easily.

6.1.3 YouTube Insight Module

This part tracks the general type of videos the user watches. It mainly checks the title, duration and the channel. After some time, it gets an idea of whether the user usually prefers tutorials, lectures, or other types of content. Later, when someone asks something like "show me the video I watched before starting work on my ML assignment", this information becomes useful.

6.1.4 Unified Query Processor

The component is considered to be the main processor of Echo. When a user query is received, this module identifies the intent of the user, i.e. whether the user wants to access specific information, get a brief overview or access interrelations among the activities that are not closely related. Instead of using precise matching of keywords, it is doing semantic interpretation, obtaining relevant information in all subordinate modules then, producing a response in a more natural speech form.

6.1.5 Analytics and Digital Well-Being Module

In addition to the ability to retrieve information, Echo is aimed at offering the users a crude audit of their time on the internet over time. The system classifies the actions of the users into such domains: study-related work, entertainment videos, professional correspondence, and the other types of activities. Such classification will give users a chance to analyze the regularity of their daily digital activities in relation to their desired goals.

6.1.6 Privacy and Permission Manager

Since Echo is handling many users' personal data, we can not ignore giving privacy to the users. Users are free to decide what information should be shared or kept on the device itself. Any type of external access only happens if the user turns it on — nothing is enabled automatically.

6.1.7 Multi-Device Synchronisation

Some users might use more than one system during the day. In that case, this part helps maintain basic consistency. Only the minimum essential information is synced, and that too only if the user allows it.

6.2 Technologies and Tools Used

The choice of these technologies will be informed by their fitting into the methodology used to develop Echo and their future stability due to the regular introduction of new features. The user interface is made with the help of HTML, CSS, and JavaScript, and React and Next.js are also integrated due to the experience of the team with these frameworks. In the case of semantic information retrieval, it will be stored in either Pinecone or Qdrant. LongChain and LlamaIndex are used to deal with handling questions and retrieving structured data. The back-end operations will be done with Node.js and Express as shown in the system- wide architecture. The Python modules (Flask or FastAPI) will be used to implement the AI-processing tasks as it will be necessary. Connections with Gmail, YouTube and Chrome APIs are established through OAuth 2.0 to guarantee permission-based access to data. The security measures are encryption and permission handling to protect data.We use Jest or Mocha for testing, while GitHub Actions helps with basic automation during development.

6.3 Expected Deliverables

- Instead of pushing all tasks together, the work is carried out in separate phases to make the development clearer and more manageable.
- Initially, research was done and the main system layout was finalised.
- Then, a prototype will be built that works with basic Chrome and Gmail usage.
- After that, the focus will shift to YouTube data support, better semantic understanding and initial well-being analysis.
- Finally, once the features are stable, interface refinement, documentation and full testing using practical examples will be completed before moving into Phase 2.

Definitions, Acronyms, and Abbreviations

1.Definitions

In the context of this research paper, Echo refers to a simple tool that allows users to re-access past internet-based actions, such as web surfing, e-mail communication, and video viewing. Instead of the users needing to repeat certain wording, Echo provides search using common questions, thus, recreating previous communication.

The system uses Retrieval Augmented Generation, where the first stage is the search of available relevant stored information and then uses an artificial intelligence model to generate the response. The method increases the applicability and sense of the output to the query.

When it comes to a setting of embedding vectors, it works by converting textual information into numbers, which makes it possible to evaluate semantic similarity between different pieces of information in Echo.

Semantic search is based on an effort to understand the intent of the user of the phrase, and not scanning through lexemes. As a result, it is still effective even when queries are put in a non-technical manner.

Digital well-being analytics are utilized to gain a crude summary of the amount of time a person spends online and the types of interactions a person has, thus, helping to understand the use patterns of a person better.

A Chrome extension, in this case, is a support tool that has been incorporated into a browser and it allows Echo to have access to authorized user information and provide the capabilities of recalls in a smooth manner.

2.Acronyms

ECHO – Efficient Contextualized History Orchestrator

RAG – Retrieval-Augmented Generation

NLP – Natural Language Processing

FAISS – Facebook AI Similarity Search

API – Application Programming Interface

LLM – Large Language Model

3.Abbreviations

JSON – JavaScript Object Notation

CSV – Comma-Separated Values

URL – Uniform Resource Locator

ID – Identifier

Conclusion

Echo is solving a major problem in digital spaces by offering to reduce the load of time burning for data recovery, needed info is scattered across multiple platforms making retrieval a big stress and causing inefficiency in work

In Phase I we highlighted major limitations in existing systems like

1. Fragmented search functions
2. Weak personalization
3. Lack of semantic search

We need to solve this because it's hard for users to recall past activity

Echo puts forward an AI powered assistance that merges Youtube Gmail and browsing history in one single vector based memory space,understanding natural language through RAG framework, the system provides assistance capable of semantic search behavior analysis and summaries, the major strengths are:

1. Multi-platform data fusion
2. Native browser integration
3. Privacy-focused local processing
4. Digital wellbeing and behavior analysis

Echo is fully feasible through existing APIs, open source LLMs vector databases and chrome extension features. In Phase II we will begin with

1. Building our RAG architecture
2. Work on chrome extension interface
3. Improving data retrieval and generation
4. Evaluating performance

Echo is making a huge impact in the digital world aiming to increase productivity and data retrieval efficiency , reducing info overwhelming on humans and improving their digital life

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